The Journal of Chemical Physics

An accurate and simple quantum model for liquid water

Cite as: J. Chem. Phys. 125, 184507 (2006); https://doi.org/10.1063/1.2386157
Submitted: 02 August 2006 • Accepted: 09 October 2006 • Published Online: 13 November 2006

Francesco Paesani, Wei Zhang, David A. Case, et al.

An accurate and simple quantum model for liquid water

Francesco Paesani
Center for Biophysical Modeling and Simulation, University of Utah, 315 South 1400 East Room 2020, Salt Lake City, Utah 84112 and Department of Chemistry, University of Utah, 315 South 1400 East Room 2020, Salt Lake City, Utah 84112

Wei Zhang and David A. Case
Department of Molecular Biology, The Scripps Research Institute, 10550 North Torrey Pines Road, TPC15, La Jolla, California 92037

Thomas E. Cheatham III
Department of Medical Chemistry, University of Utah, 2000 East 30 South, Skaggs Hall 201, Salt Lake City, Utah 84112; Department of Pharmaceutics, University of Utah, 2000 East 30 South, Skaggs Hall 201, Salt Lake City, Utah 84112; and Department of Pharmaceutical Chemistry and Bioengineering, University of Utah, 2000 East 30 South, Skaggs Hall 201, Salt Lake City, Utah 84112

Gregory A. Voth[a]
Center for Biophysical Modeling and Simulation, University of Utah, 315 South 1400 East Room 2020, Salt Lake City, Utah 84112 and Department of Chemistry, University of Utah, 315 South 1400 East Room 2020, Salt Lake City, Utah 84112

(Received 2 August 2006; accepted 9 October 2006; published online 13 November 2006)

The path-integral molecular dynamics and centroid molecular dynamics methods have been applied to investigate the behavior of liquid water at ambient conditions starting from a recently developed simple point charge/flexible (SPC/Fw) model. Several quantum structural, thermodynamic, and dynamical properties have been computed and compared to the corresponding classical values, as well as to the available experimental data. The path-integral molecular dynamics simulations show that the inclusion of quantum effects results in a less structured liquid with a reduced amount of hydrogen bonding in comparison to its classical analog. The nuclear quantization also leads to a smaller dielectric constant and a larger diffusion coefficient relative to the corresponding classical values. Collective and single molecule time correlation functions show a faster decay than their classical counterparts. Good agreement with the experimental measurements in the low-frequency region is obtained for the quantum infrared spectrum, which also shows a higher intensity and a redshift relative to its classical analog. A modification of the original parametrization of the SPC/Fw model is suggested and tested in order to construct an accurate quantum model, called q-SPC/Fw, for liquid water. The quantum results for several thermodynamic and dynamical properties computed with the new model are shown to be in a significantly better agreement with the experimental data. Finally, a force-matching approach was applied to the q-SPC/Fw model to derive an effective quantum force field for liquid water in which the effects due to the nuclear quantization are explicitly distinguished from those due to the underlying molecular interactions. Thermodynamic and dynamical properties computed using standard classical simulations with this effective quantum potential are found in excellent agreement with those obtained from significantly more computationally demanding full centroid molecular dynamics simulations. The present results suggest that the inclusion of nuclear quantum effects into an empirical model for water enhances the ability of such model to faithfully represent experimental data, presumably through an increased ability of the model itself to capture realistic physical effects.
[DOI: 10.1063/1.2386157]

I. INTRODUCTION

Since liquid water plays a fundamental role in many natural phenomena, much attention has been devoted in the past to its modeling at the atomic scale. As a result, a large number of different water models have been developed, from very simple ones that assume rigid geometries and fixed point charge distributions[1,2] to other more sophisticated forms that include flexibility[3,4] and polarization effects.[5–7] These models are routinely applied in classical simulations to investigate either the properties of bulk water or its behavior as a solvent. However, given the low mass of hydrogen atoms, classical mechanics is often not accurate enough to describe the liquid phase even at ambient conditions.[8] It is well known, for instance, that the intramolecular vibrations behave quantum mechanically even at quite high temperatures, and that some of the intermolecular vibrational modes, such as the librational motions, are significantly quantum mechanical at room temperature.[8–10]

[a]Electronic mail: voth@chem.utah.edu

However, contrary to their classical analogs, quantum simulations of liquid water are rare. The first quantum calculations were performed in the mid-1980s, using the path-integral Monte Carlo (PIMC) method with a rigid water model to analyze the impact of quantum fluctuations on the liquid structure.[8,11] Later, the effects of nuclear quantization on the liquid dynamics were also investigated. The centroid molecular dynamics (CMD) method was applied to study the behavior of a flexible model of water at ambient conditions.[9] The less sophisticated Feynman-Hibbs effective potential approach was then employed to compute the self-diffusion coefficient of light (H_2O) and heavy (D_2O) water over a rather wide temperature range.[12] The CMD method was also used with a rigid-body water model in several studies on the orientational and translational motions.[13] More recently the static and dynamical properties of a flexible water model have been computed using the Feynman-Kleinert linearized path-integral method,[10] while the ring polymer molecular dynamics approach with a rigid-body water model has been applied to the study of quantum diffusion and relaxation processes.[14] Quantum properties of the liquid were also computed using path-integral molecular dynamics (PIMD) simulations with two different polarizable water models.[15,16] Since, with the exception of Refs. 15 and 16 the water models employed in the quantum simulations were originally parametrized using classical mechanics, all these studies demonstrated that the agreement between quantum and experimental results is in general not satisfactory. In particular, it was shown that inclusion of nuclear quantization leads to a faster diffusion than that observed in purely classical simulations. Depending on the interaction potential and approximate quantum dynamical method employed, the quantum self-diffusion coefficient is found to be $\sim$1.5–2.0 larger than its classical counterpart. This is consistent with the decrease in the structure of the liquid due to the disruption of the hydrogen bonding network that was observed in the earlier PIMC calculations.[8,11] Since nature intrinsically obeys quantum mechanics, the conclusions of these studies suggest that the development of a water model based on purely quantum calculations would be not only desirable but should also provide a more accurate and realistic description of the liquid properties at ambient conditions. Furthermore, considering that quantum simulations of biomolecules are now possible,[17,18] the new water model should be simple in order to be interfaced with the relatively simple force fields generally employed to describe protein interactions, as well as fast to compute since in a typical biomolecular simulation the atoms of the water molecules represent about 80% of the total number of particles.

As mentioned above, the use of quantum simulations to investigate the properties of many-body systems such as liquid water has not been common so far. The main reason for this has been represented by the associated computational cost that becomes quite high even for systems with a few hundreds of particles. Among the approximate quantum methods that have been developed (e.g., centroid molecular dynamics,[19–21] quantum mode coupling theory,[22] forward-backward semiclassical dynamics,[23] and Feynman-Kleinert linearized path-integral method[24]), the CMD method apparently accounts for many of the principal quantum features in a many-body system. Nevertheless, the evaluation of the centroid force necessary to propagate the centroid variables according to classical-like equations of motion requires significant computational effort that makes this method essentially inapplicable to large systems with thousands of degrees of freedom. It is thus evident that if a faster numerical procedure to compute the centroid force would be devised, then the CMD method could be applied to a much wide range of problems. A few attempts have been made in the past along this direction.[25] Recently a force-matching approach has been proposed and applied with success to the computation of quantum dynamical properties of homogeneous systems such as liquid *para*-hydrogen and *ortho*-deuterium.[26] Based on the predetermination of the centroid force, this methodology greatly reduces the computational time, which indeed becomes comparable to that required by a purely classical simulation. At present, this force-matching scheme appears to be a promising method to perform (effective) quantum simulations even of complex many-body systems.

para-hydrogen:
2 protons have
parallel spin
=> total spin =0

ortho-deterium:
1 proton and 1
neutron have
parallel spin
=> total spin =1

The present study thus aims at the following: (1) developing a simple and accurate quantum model for water that is capable of accurately describing both equilibrium and dynamical properties of the liquid at ambient conditions which could be eventually used in quantum biomolecular simulations, and (2) constructing a corresponding effective quantum model that can be used in standard classical simulations to obtain a fully quantum description of the liquid. In order to accomplish these two goals PIMD (Ref. 27) and CMD simulations were first performed with the recently developed SPC/Fw water model.[28] Several equilibrium and dynamical properties of the liquid at ambient conditions were computed. Since a relatively poor agreement between the calculated and experimental values was found, the original parameters of the SPC/Fw model were then modified based exclusively on quantum results until an optimal set was obtained. This procedure leads to a new and fully quantum model for water, the q-SPC/Fw model. When utilized in quantum simulations, it is shown that this new model provides an accurate description of several liquid properties at ambient conditions. A force-matching approach within the CMD framework was subsequently applied to the q-SPC/Fw force field in order to generate an effective quantum model in which the effects of the nuclear quantization are explicitly distinguished from those due to the underlying molecular interactions. It is shown that when employed in standard classical simulations this effective quantum force field allows one to reproduce with considerable accuracy the quantum results obtained from the significantly more computationally demanding full PIMD and CMD simulations performed with the pure q-SPC/Fw model.

The paper is organized as follows: In Sec. II A, the computational and numerical details specific to the quantum simulations are provided, while the quantum results obtained with both the original SPC/Fw and new quantum q-SPC/Fw water models are discussed in Sec. II B in light of the corresponding classical and experimental values. The force-matching procedure within the CMD framework is then

TABLE I. Parameters of the original SPC/Fw and new quantum q-SPCF/Fw water models considered in this study. The functional from of the potential is defined in Eqs. (2) and (3).

Model	k_b (kcal mol^{-1} Å^{-2})	$r_{\mathrm{OH}}^{\mathrm{eq}}$ (Å)	k_a (kcal mol^{-1} rad^{-2})	$\vartheta_{\angle\mathrm{HOH}}^{\mathrm{eg}}$ (deg)	$\varepsilon_{\mathrm{OO}}$ (kcal mol^{-1})	σ_{OO} (Å)	q_{O} (e)	q_{H} (e)
SPC/Fw	1059.162	1.012	75.90	113.24	0.155 425 3	3.165 492	−0.82	0.41
q-SPC/FW	1059.162	1.000	75.90	112.0	0.155 425 2	3.165 492	−0.84	0.42

briefly reviewed in Sec. III A, and the results obtained from its application to the q-SPC/Fw model are presented in Sec. III B. The conclusions are given in Sec. IV.

II. QUANTUM SIMULATIONS

A. Computational methods and numerical details

The equilibrium properties of liquid water at ambient conditions ($T=298.15$ K and $P=1$ atm) were computed using the path integral–molecular dynamics method in the normal mode representation (NMPIMD).[27] The quantum dynamical simulations were performed using the CMD method[19–21] with the centroid force computed "on the fly" according to the adiabatic time scale separation scheme.[20] Since both methods have been exhaustively described in the literature, only the computational details that are specific to the present study are given here.

Both NMPIMD and CMD methods have been implemented within the AMBER molecular dynamics package,[17] which was then employed in all calculations. The simulations were carried out for a system of $N=216$ water molecules with simple cubic periodic boundary conditions. The short-range interactions were truncated at an atom-atom distance of 9.0 Å, while the electrostatic interactions were treated by using the particle mesh Ewald method with 5832 grid points and a precision of 10^{-6}.[29] Initial classical molecular dynamics runs for a total of 500 ps were performed with temperature and pressure maintained via Langevin dynamics and Berendsen barostat,[30] respectively. A NMPIMD simulation was then carried out in the *NPT* ensemble for 1 ns during which both the density and the enthalpy of vaporization were computed. Canonical equilibrium properties were instead obtained from a NMPIMD run of 1 ns in the *NVT* ensemble. In all quantum calculations the propagation was performed using the velocity-Verlet algorithm[31] with a time step $\Delta t=0.1$ fs. The temperature was controlled via Nosé-Hoover chains (NHCs) of four thermostats[32] that were coupled to each path-integral normal mode. The isothermal-isobaric ensemble was generated according to the algorithm illustrated in Ref. 33. Analysis of both energetics and structural properties indicated that a discretization of the quantum partition function with 24 quasiparticles or beads[27] was sufficient to obtain converged results, and therefore it was employed in all NMPIMD and subsequent CMD calculations.

In the CMD simulations the propagation of the centroid degrees of freedom was carried out in the *NVE* ensemble with NHCs of four thermostats attached to each nonzero frequency normal mode. An adiabaticity parameter[20] $\gamma=0.025$ and a time step $\Delta t=0.005$ fs ensured a sufficiently large separation in time between the motion of the centroid and the other nonzero frequency normal modes, as well as a proper integration of the latter. Several time correlation functions describing both collective and individual molecular relaxation processes were computed by averaging over 80 independent centroid trajectories of 10 ps each. Running several relatively short trajectories was preferred over performing fewer but longer simulations in order to guarantee a proper canonical average of the initial centroid variables. As discussed in detail elsewhere,[21] a time correlation function computed by means of CMD represents an approximation to the corresponding Kubo-transformed correlation function $C_{AB}^{(K)}(t)=\langle\hat{A}^{(K)}(0)\hat{B}^{(K)}(t)\rangle$, where $\hat{A}$ and $\hat{B}$ are two operators describing any physical observable. If $\hat{A}$ is a linear function of the position and/or the momentum operators, $C_{AB}^{(K)}(t)$ is related to the exact quantum correlation function through the following relationship:

$$\tilde{C}(\omega)=\frac{\hbar\beta\omega}{2}\left[\coth\left(\frac{\hbar\beta\omega}{2}\right)+1\right]\tilde{C}^{(K)}(\omega). \quad (1)$$

In Eq. (1), $\tilde{C}^{(K)}(\omega)$ and $\tilde{C}(\omega)$ are the Fourier transform of the Kubo transformed and exact quantum correlation function, respectively. For a nonlinear operator $\hat{A}$, it has been shown that the centroid correlation function still remains a well-defined approximation to the corresponding Kubo-transformed correlation function.[21,34] However, the relationship between higher order Kubo-transformed correlation functions and the corresponding exact quantum correlation functions is not as simple as in the linear case.[34]

The first set of NMPIMD and CMD calculations was performed using the SPC/Fw water force field[28] that takes the form

$$V^{\mathrm{intra}}=\frac{k_b}{2}[(r_{\mathrm{OH}_1}-r_{\mathrm{OH}}^{\mathrm{eq}})^2+(r_{\mathrm{OH}_2}-r_{\mathrm{OH}}^{\mathrm{eq}})^2]+\frac{K_a}{2}(\vartheta_{\angle\mathrm{HOH}}-\vartheta_{\angle\mathrm{HOH}}^{\mathrm{eq}})^2, \quad (2)$$

$$V^{\mathrm{inter}}=\sum_{i,j}^{\text{all pairs}}\left\{4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{R_{ij}}\right)^{12}-\left(\frac{\sigma_{ij}}{R_{ij}}\right)^{6}\right]+\frac{q_iq_j}{R_{ij}}\right\}. \quad (3)$$

Here, V^{intra} and V^{inter} are the intra- and intermolecular interactions; $r_{\mathrm{OH}}^{\mathrm{eq}}$ and $\vartheta_{\angle\mathrm{HOH}}^{\mathrm{eq}}$ are the equilibrium bond length and angle of an isolated water molecule; R_{ij} is the distance between atoms i and j; ε_{ij} and σ_{ij} are the Lennard-Jones parameters for atom pair (i,j); and q_i is the partial charge on atom i. The values of all the parameters entering the SPC/Fw model are listed in Table I. It has recently been shown that when employed in classical simulations this three-site, flexible model accurately describes many equilibrium and dy-

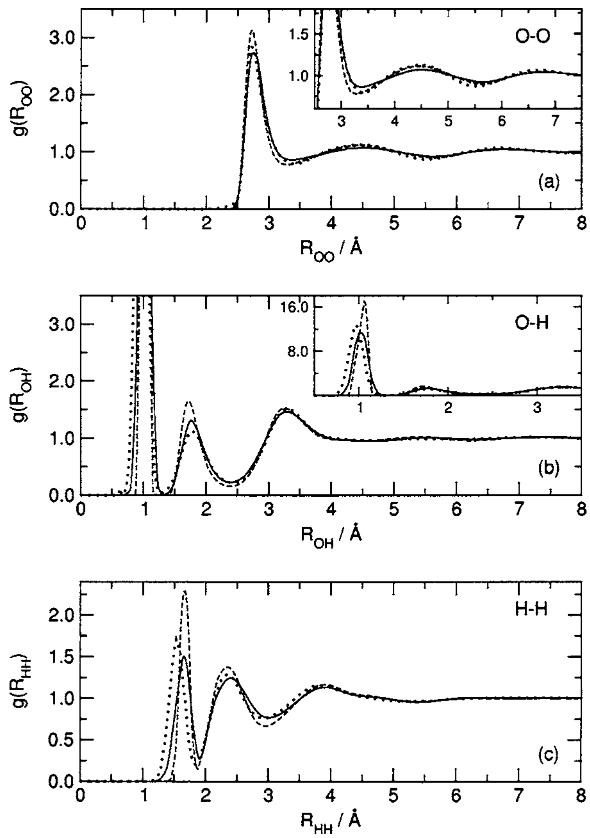

FIG. 1. Radial distribution functions for oxygen-oxygen (a) (the inset is a magnification of the second and third peak), oxygen-hydrogen (b), and hydrogen-hydrogen (c) atom pairs computed with the SPC/Fw water model. Solid line: quantum results from normal-mode path-integral molecular dynamics simulations. Dashed line: results from classical molecular simulations (Ref. 28). Dotted line: experimental data from Ref. 36, panel (a), and from Ref. 35, panels (b) and (c).

namical properties of liquid water at $T=298.15$ K and $P=1$ atm.[28]

Based exclusively on the quantum results obtained from NMPIMD and CMD simulations, a fully quantum model of water was then developed (see Table I) by tuning the parameters of the original SPC/Fw force field until an overall good agreement between the computed and the available experimental values was achieved for a large set of equilibrium and dynamical properties of liquid water at ambient conditions.

B. Results

The radial distribution functions (RDF's) computed with the quantized SPC/Fw model using the NMPIMD method are compared in Fig. 1 with the corresponding classical[28] and experimental[35,36] results. As found for other water models,[8,9,11–13] the quantum oxygen-oxygen RDF from the SPC/Fw force field (top panel of Fig. 1) appears to be less structured than its classical analog. This is a consequence of the inclusion of the nuclear quantization that results in a weaker hydrogen bond network. Comparison with the experimental data shows that the quantum RDF reproduces with better accuracy both the height and the position of the first peak at ~2.7 Å, while it appears to be again slightly less structured for larger O–O separations. Similar behavior is also found in the oxygen-hydrogen (middle panel of Fig. 1) and hydrogen-hydrogen (bottom panel of Fig. 1) radial distribution functions with the quantum results always being less structured than their classical counterparts. For both these RDF's, inclusion of quantum effects improves the agreement with the experimental data. This is particularly evident for the first peaks describing the intramolecular O–H (middle panel, see also insert) and H–H (bottom panel) vibrations. It is also interesting to note that in the quantum results the second peak (at ~1.7 Å) in the O–H radial distribution function becomes lower than the third one at ~3.3 Å. This is consistent with the experimental data and suggests that the longer lifetime of the hydrogen bonds obtained from a classical simulation results in an anomalously stable hydrogen bond network.

The impact of nuclear quantization on the equilibrium properties of liquid water at ambient conditions was further analyzed by computing the density (ρ), the enthalpy of vaporization (ΔH_{vap}), the dielectric constant (ε), and the heat capacity at constant volume (C_v) using the NMPIMD method. Following Ref. 2 the enthalpy of vaporization was approximated as

$$\Delta H_{\mathrm{vap}}(T) = E_{\mathrm{gas}} - E_{\mathrm{liq}} + P(V_{\mathrm{gas}} - V_{\mathrm{liq}}) \approx \langle E_{\mathrm{gas}}\rangle - \langle E_{\mathrm{liq}}\rangle + RT - P\langle V_{\mathrm{liq}}\rangle + \xi, \tag{4}$$

where $\langle E_{\mathrm{gas}}\rangle$ and $\langle E_{\mathrm{liq}}\rangle$ are the average quantum energies for N water molecules in the gas and liquid phases, respectively, and $\langle V_{\mathrm{liq}}\rangle$ is the average volume of the simulation box, which were computed during a 1 ns *NPT* simulation carried out at $T=298.15$ K and $P=1$ atm. In Eq. (4) R is the gas constant and ξ corrects for the approximations made in the simulation and in the derivation of the expression for ΔH_{vap}. For a quantum simulation this term accounts for polarization, nonideal gas, and pressure effects. All these contributions are described in detail in Ref. 2.

The dielectric constant was calculated as

$$\varepsilon = 1 + \frac{4\pi}{3Vk_BT}(\langle \mathbf{M}^2\rangle - \langle \mathbf{M}\rangle^2), \tag{5}$$

where $\mathbf{M}$ is the total dipole moment of the system computed using a 1 ns *NVT* simulation, V is the volume of the simulation box, and k_B is the Boltzmann's constant.

For the heat capacity at constant volume several estimators that show smaller statistical uncertainties have been devised for path-integral calculations.[37] However, they are computationally expensive and, for the purposes of the present study, a simple finite difference formula was found sufficient to assess the effect of nuclear quantization. Therefore, the heat capacity at constant volume was obtained as

De novo contornaram o cálculo das integrais de caminho por causa do custo computacional

$$C_V(T) = \frac{E(T+\Delta T) - E(T-\Delta T)}{2\Delta T}, \tag{6}$$

where $E(T)$ is the internal energy of the system expressed according to the path-integral virial estimator.[38] In order to compute $C_V(T)$ at $T=298.15$ K two additional NMPIMD simulations were thus performed at $T=288.15$ K and $T=308.15$ K, respectively.

The present quantum results for ρ, ΔH_{vap}, ε, and C_V are compared with the their classical counterparts and corre-

TABLE II. Comparison between experimental (second column) and calculated values for several equilibrium and dynamical properties of liquid water at ambient conditions. The quantum results from normal-mode path-integral and centroid molecular dynamics simulations with the SPC/Fw model are listed in the third column, while the corresponding classical results from Ref. 28 are listed in the fourth column. The NMPIMD and CMD results obtained with the new q-SPC/Fw model are listed in the last column. The experimental results are taken from Table II of Ref. 28. The statistical uncertainties are given as standard deviations.

	Expt.	Classical SPC/Fw	Quantum SPC/Fw	Quantum q-SPC/Fw
ρ	0.997	1.01 ± 0.02	0.991 ± 0.005	0.999 ± 0.005
ΔH_{vap} (kcal mol^{-1})	10.52	11.2 ± 0.2[a]	9.4 ± 0.2	9.6 ± 0.2
C_v (cal mol^{-1} K^{-1})	17.7	27 ± 1	21 ± 1	22 ± 1
ε	78.5	80 ± 2	64 ± 4	86 ± 4
D (Å ps^{-1})	0.23	0.232 ± 0.005	0.32 ± 0.01	0.24 ± 0.01
τ_D	8.3	9.50 ± 0.1	7.2 ± 0.5	8.5 ± 0.5
τ_1^{HH} (ps)	⋯	3.90 ± 0.1	3.0 ± 0.1	3.8 ± 0.1
τ_2^{HH} (ps)	2.0	2.01 ± 0.1	1.35 ± 0.05	1.85 ± 0.05
τ_1^{OH} (ps)	⋯	4.17 ± 0.1	3.1 ± 0.1	4.1 ± 0.1
τ_2^{OH} (ps)	1.95	1.86 ± 0.1	1.25 ± 0.05	1.70 ± 0.05
τ_1^{μ} (ps)	⋯	4.70 ± 0.1	3.3 ± 0.1	4.7 ± 0.1
τ_1^{μ} (ps)	1.9	1.72 ± 0.1	1.15 ± 0.05	1.60 ± 0.05

[a]ΔH_{rap} from Ref. 28 has been corrected according to Eq. (4) with the inclusion of the quantum correction (see Ref. 2). The purely classical value of ΔH_{vap} is 10.4 Kcal/mol.

sponding experimental data in Table II. For all these quantities the quantum values are smaller than their classical counterparts as expected for a less structured liquid with a weaker hydrogen bonding network. In particular, the lower density obtained in the case of the quantum liquid is a direct manifestation of the atomic delocalization resulting from the inclusion of nuclear quantization. This evidently increases the effective diameter of the water molecules and consequently their molar volume. It is also interesting to note that the quantum results for the heat capacity at constant volume are in better agreement with the experimental values than their classical counterparts, while ε is significantly underestimated. All of these results taken together for the quantized SPC/Fw model suggested the necessity for its reparametrization for quantum simulations, the results of which are described later.

In order to analyze the effect of nuclear quantization on the dynamical properties of liquid water, the self-diffusion coefficient was computed using the well-known expression[31]

$$D=\frac{1}{3}\int_0^{\infty}\langle\mathbf{v}(t)\cdot\mathbf{v}(0)\rangle=\frac{1}{3}\int_0^{\infty}C_{vv}(t)dt, \tag{7}$$

where $C_{vv}(t)$ is the quantum velocity autocorrelation function. This was obtained from the corresponding centroid correlation function via Eq. (1). The self-diffusion coefficient can also be computed from the slope of the mean-square displacement, $\langle\Delta\mathbf{R}(t)^2\rangle$, as a function of time according to the Einstein equation[31]

$$D=\lim_{t\to\infty}\frac{\langle|\mathbf{R}(t)-\mathbf{R}(0)|^2\rangle}{6t}=\lim_{t\to\infty}\frac{\langle\Delta\mathbf{R}(t)^2\rangle}{6t}. \tag{8}$$

Both centroid and classical velocity autocorrelation functions are shown in the left panel of Fig. 2, while a comparison between the corresponding mean-square displacements is shown in the right panel of the same figure. The computed and experimental values for the self-diffusion coefficient are reported in Table II. Inclusion of the quantum effects increases D by about 50%. Similar results were also obtained with different water models.[9,10,13,14] This increase can be explained by considering that nuclear quantization effectively weakens the interactions between molecules with the consequence of lowering the barriers for the activated molecular diffusion. Seria relevante no estudo de superfluxo da água?

The effect of nuclear quantization on the orientational relaxation dynamics was also investigated by computing

$$\Phi(t)=\frac{\langle\mathbf{M}(0)\mathbf{M}(t)\rangle-\langle\mathbf{M}\rangle^2}{\langle\mathbf{M}^2(0)\rangle-\langle\mathbf{M}\rangle^2}, \tag{9}$$

where the total dipole moment quantum autocorrelation function $\langle\mathbf{M}(0)\mathbf{M}(t)\rangle$ was obtained from the corresponding centroid quantity via Eq. (1). The relaxation time constant of $\Phi(t)$, which represents the Debye dielectric relaxation time,[39] was computed by numerical integration assuming an exponential decay of $\Phi(t)$ at long times, $\tau_D=\int_0^{-\infty}\Phi(t)dt$. The quantum result for τ_D is compared with the corresponding classical and experimental values in Table II.

Typical single molecule relaxation processes were examined by calculating autocorrelation functions of the type

$$C_l(t)=\langle P_l[\mathbf{e}(t)\cdot\mathbf{e}(0)]\rangle, \tag{10}$$

where P_l is the Legendre polynomial of order l and $\mathbf{e}$ is a unit vector along a specific axis in the body-fixed reference frame of each water molecule. As discussed in Sec. II A, for correlation functions involving nonlinear operators of position and momentum such as the orientational correlation functions defined in Eq. (10), there is no simple relationship between the centroid and the exact quantum values. Therefore, the first- and second-order orientational relaxation times τ_l were computed by integrating directly the corresponding centroid autocorrelation functions after analytically fitting their exponential tails. The comparison between the centroid and the classical values for several relaxation times (Table II) indi-

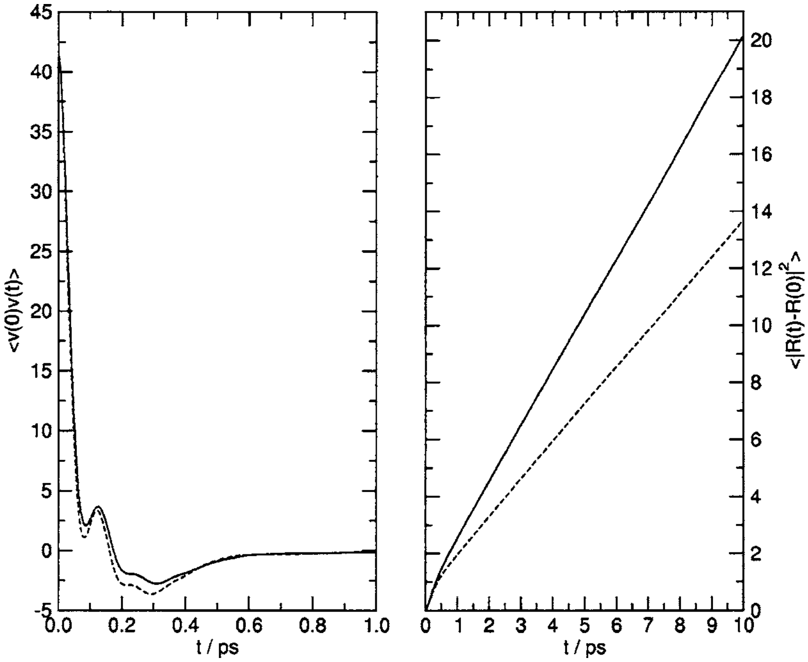

FIG. 2. Velocity autocorrelation function (left panel) and mean-square displacement (right panel) of liquid water computed with the SPC/Fw model. Solid line: results from centroid molecular dynamics simulations; dashed line: results from classical molecular dynamics simulations.

cates that the effect of nuclear quantization is large, resulting in significantly faster relaxation dynamics. This more rapid exponential decay of the CMD correlation functions is clearly evident in Fig. 3, which shows a comparison between the classical and CMD results for $C_l^{(\mu)}(t)$ that was computed by defining the unit vector $\mathbf{e}$ of Eq. (9) along the direction of the molecular dipole moment $\boldsymbol{\mu}$. Both classical and CMD functions display an initial fast decorellation for $t \leqslant 35$ ps corresponding to a dynamical regime where the water molecules rotate freely. Due to the molecular collisions, the orientational dynamics then enters a diffusive regime that is responsible for the slower decay of $C_l^{(\mu)}(t)$ seen at longer t.

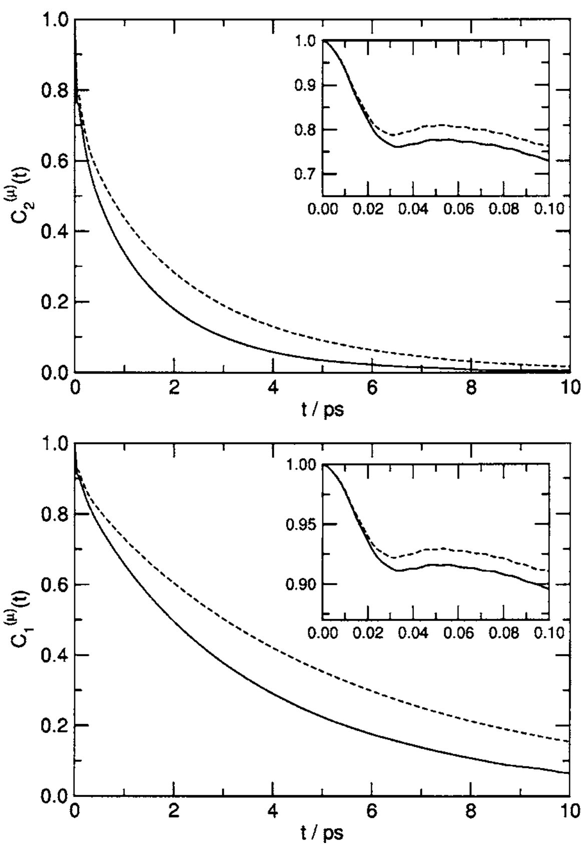

FIG. 3. Orientational correlation function $C_l^{(\mu)}(t)$, Eq. (10), with $l=1$ (bottom panel) and $l=2$ (upper panel) computed with the SPC/Fw model. Solid line: results from centroid molecular dynamics simulations; dashed line: results from classical molecular dynamics simulations.

The infrared absorption coefficient $\alpha(\omega)$ was computed from the total dipole moment quantum autocorrelation function as[40]

$$\alpha(\omega) = \left[\frac{4\pi^2\omega}{3V\hbar c n(\omega)}\right](1-e^{-\beta\hbar\omega})\frac{1}{2\pi} \times \int_{-\infty}^{+\infty} e^{-i\omega t}\langle \mathbf{M}(0)\mathbf{M}(t)\rangle dt, \tag{11}$$

where $n(\omega)$ is the refractive index of the system at frequency ω, and c is the speed of light. As found in previous studies,[9,10] the quantum spectrum shown in Fig. 4 is red-shifted and has considerably higher intensity than its classical analog. The frequency shift is large for both the symmetric (at $\sim$3420 cm^{-1}) and asymmetric (at $\sim$3530 cm^{-1}) O–H stretches, while becomes smaller for the H–O–H bend at $\sim$1450 cm^{-1}, and for the broadband corresponding to the hindered motions ($\sim$250–900 cm^{-1}).[41] Although the frequencies corresponding to the intramolecular vibrations are in reasonable agreement with the experimental values with the largest difference ($\sim$200 cm^{-1}) being found for the bending vibration, the shape of these peaks is not well reproduced. This is particularly evident in the case of the O–H stretches for which the SPC/Fw model predicts two separated peaks instead of the broadband (due to the spectral diffusion) observed in the experimental spectrum. This difference between theory and experiment is not surprising since the intramolecular vibrations in the SPC/Fw model are described as simple uncoupled harmonic oscillators. By contrast, very good agreement between the shapes of quantum

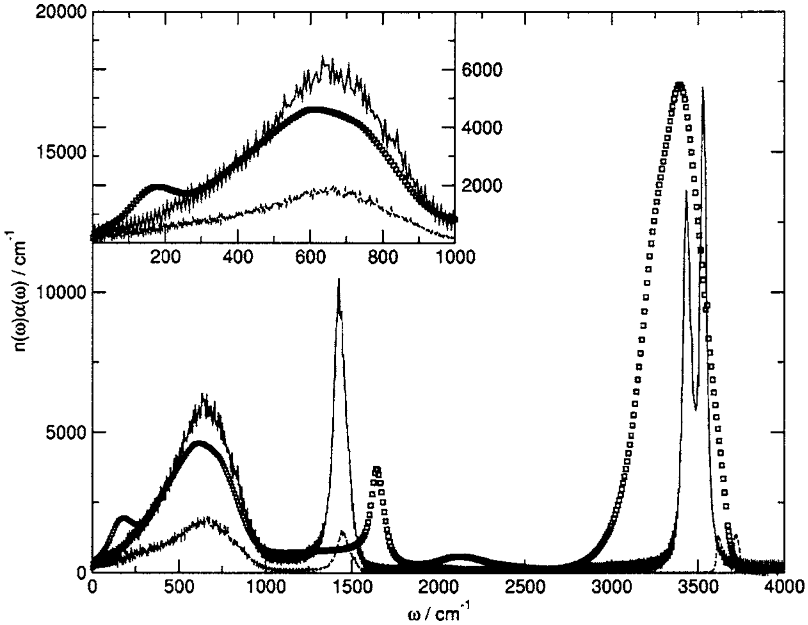

FIG. 4. Infrared spectrum of liquid water at $T = 298.15$ K. The inset is a magnification of the low-frequency region. Solid line: quantum results obtained via Eq. (1) from centroid molecular dynamics simulations; dashed line: results from classical molecular dynamics simulation; open squares: experimental data from Ref. 42. The theoretical spectra were computed with the SPC/Fw model.

and experimental[42] spectra is found for the broadband describing hindered motions occurring in the far infrared region (inset of Fig. 4). Note that the appearance in the experimental spectrum of the small shoulder at $\sim$200 cm^{-1} results from a dipole-induced-dipole mechanism,[43] and consequently it cannot be reproduced by any water model that has a fixed point charge distribution as the SPC/Fw force field employed here.

As stated earlier, from the detailed analysis on the properties of liquid water presented above it is clear that the agreement between the experimental values and the quantum results obtained with the SPC/Fw model is not completely satisfactory especially for the dynamical properties such as the self-diffusion coefficient and the relaxation times. This is not unexpected since the SPC/Fw water model, as most of the other models discussed in the literature, was explicitly parametrized to reproduce experimental observables employing only classical simulations.[28]

In order to correct for this behavior, and more importantly to derive a water model suitable for quantum simulations, the SPC/Fw force field was modified by tuning some of its parameters. Since it was found that quantum calculations provided a less structured liquid for which the dielectric constant was underestimated, and both the translational and rotational dynamics were significantly faster, as a first step it was decided to increase the negative (positive) partial charge on the oxygen atom (hydrogen atoms) from −0.82 (+0.41) to −0.85 (+0.425), in increments of 0.1 (0.05). Although, a choice of −0.85 (+0.425) provided an excellent agreement between quantum and experimental values for several properties, the resulting value for the dielectric constant was found to be exceedingly large. The oxygen (hydrogen) partial charge was therefore fixed to a value of −0.84 (+0.42). Then, by considering that the inclusion of the nuclear quantization effectively increases the dimensions of the water molecules, a further refinement of the model was performed by slightly reducing the values of both the equilibrium bond distance, $r_{\mathrm{OH}}^{\mathrm{eq}}$, and angle, $\vartheta_{\angle\mathrm{HOH}}^{\mathrm{eq}}$, of Eq. (2). Given the fairly good agreement between the computed and the experimental spectrum, no attempt was made to further optimize the force constants describing the intramolecular vibrations, which would require a more sophisticated modeling[3] that is beyond the purpose of the present study. The optimized parameters for the new quantum model q-SPC/Fw are listed in Table I.

The quantum radial distribution functions computed with the new q-SPC/Fw model are shown in Fig. 5. All three RDF's are slightly more structured than the corresponding quantum distributions obtained with the original SPC/Fw. The quantum values computed for several equilibrium and dynamical properties with the new q-SPC/Fw model are also reported in the last column of Table II. Comparison with the corresponding results obtained with the quantized version of the original SPC/Fw model (third column of Table II) clearly shows a significantly better agreement with the experimental data. The quantum infrared spectrum computed with the q-SPC/Fw model (not shown) is only slightly blueshifted ($\sim$10 cm^{-1}) in comparison with the corresponding spectrum obtained with the original SPC/Fw model and, consequently, it is undistinguishable from the latter on the scale of Fig. 4. This analysis on the equilibrium and dynamical water properties indicates that, when employed in quantum simulations, the new q-SPC/Fw model provides an accurate description of the liquid phase at ambient conditions.

As mentioned in the Introduction, most of the quantum simulations of liquid water reported in the literature have been performed employing empirical models that, to some degree, already incorporate nuclear quantum effects implicitly. Therefore, a direct and fair comparison with the quantum results obtained here with the q-SPC/Fw force field is not possible since all those models (once quantized) provide a very poor description of the liquid properties. It is instead possible as well as interesting to compare the present quantum results with those obtained in Refs. 15 and 16 using PIMD simulations with the MCDHO (Ref. 5) and QMPFF2 (Ref. 7) water models, respectively. Both these polarizable models were developed from high-level *ab initio* calculations

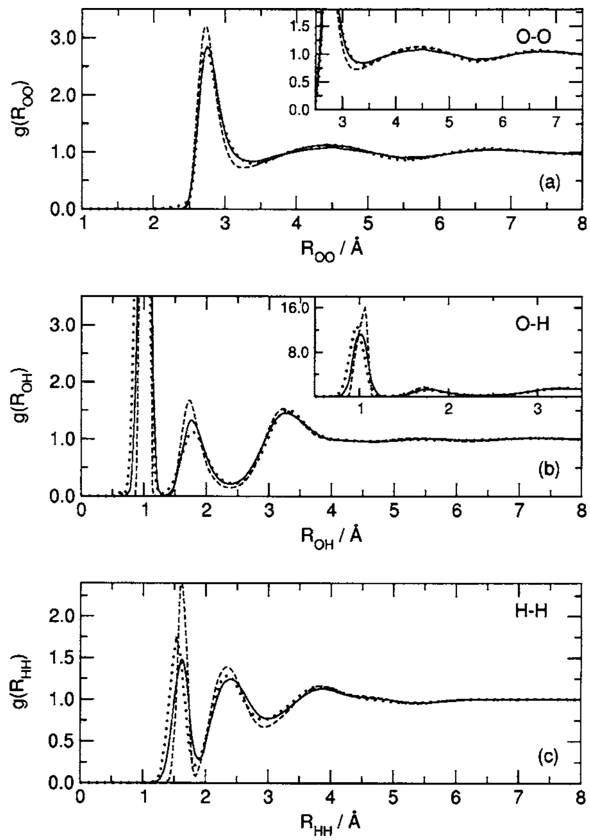

FIG. 5. Radial distribution functions for oxygen-oxygen (a) (the inset is a magnification of the second and third peak), oxygen-hydrogen (b), and hydrogen-hydrogen (c) atom pairs computed with the q-SPC/Fw water model. Solid line: quantum results from normal-mode path-integral molecular dynamics simulations. Dashed line: results from classical molecular dynamics simulations. Dotted line: experimental data from Ref. 36 (a), and from Ref. 35 [(b) and (c)].

and, consequently, they do not implicitly include any nuclear quantum effect. A comparison between the quantum radial distribution functions computed here with the q-SPC/Fw model (Fig. 5) and in Ref. 15 with the MCDHO model (Figs. 8–10 of that reference) clearly demonstrates that the q-SPC/Fw model provides a significantly better agreement with the available experimental data. In particular, the MCDHO model predicts much more structured distributions at large interatomic separations and greatly overestimates the height of all peaks with the exception of that corresponding to the hydrogen bonds at $R \approx 1.8$ Å (Fig. 9 of Ref. 15). Furthermore, the quantum H–H distribution function computed with the MCDHO (Fig. 10 of Ref. 15) is also considerably shifted towards larger interatomic separations. More difficult it is the comparison with the quantum results reported in Ref. 16 for the QMPFF2 water model. In that study, in fact, a discretization of the quantum partition function with four quasiparticles (beads) was employed which is known to be not sufficient to fully capture the quantum effects for flexible water models.[9,15] A comparison of the quantum results obtained here with the q-SPC/Fw model with those reported in Ref. 16 for the radial distribution functions (Fig. 4 of Ref. 16) as well as for other water properties (Table III of Ref. 16) again indicates that the present q-SPC/Fw model provides a significantly better description of the liquid at ambient conditions.

III. FORCE MATCHING

A. Computational method and numerical details

The new q-SPC/Fw force field was chosen as a reference model for constructing an effective quantum force field for liquid water in which, similar to the Born-Oppenheimer approximation, the effects of nuclear quantization are explicitly distinguished from those due to the underlying classical molecular interactions. This was accomplished by employing a force-matching approach within the CMD framework. The force-matching methodology has already been successfully applied to derive classical force fields from *ab initio* molecular dynamics calculations,[44] and more recently it has been employed to generate an effective quantum potential for liquid *para*-hydrogen and *ortho*-deuterium.[26] The present implementation closely follows that illustrated in Ref. 26. The main difference is represented by the extension of the algorithm to a more complex system with multiple quantum atomic species and long-range electrostatic interactions such as liquid water. Within the CMD framework the centroid force $F^c(\mathbf{R}_c)$ can be written as the classical force $F^{\mathrm{cl}}(\mathbf{R}_c)$ plus a quantum deviation $\Delta F^{\mathrm{qm}}(\mathbf{R}_c)$,

$$F^c(\mathbf{R}_c) = F^{\mathrm{cl}}(\mathbf{R}_c) + \Delta F^{\mathrm{qm}}(\mathbf{R}_c), \tag{12}$$

where $\mathbf{R}_c$ is the centroid position. From Eq. (12) it follows that the knowledge of $\Delta F^{\mathrm{qm}}(\mathbf{R}_c)$ allows one to generate centroid trajectories within a purely classical framework, thus greatly reducing the computational cost associated with a full CMD simulation. A suitable expression for $\Delta F^{\mathrm{qm}}(\mathbf{R}_c)$ can be obtained by matching the quantum deviation of the centroid force to a classical form in a least-squares minimization scheme,

$$\chi^2 = \frac{1}{3N_cN_p}\sum_{k=1}^{N_c}\sum_{j=1}^{N_p}[\Delta F_{jk}^{\mathrm{ref}}(\{R_j\}_k) - \Delta F_{jk}^{\mathrm{mod}}(\{R_j\}_k, a_1, \ldots, a_m)]^2. \tag{13}$$

Here, N_c and N_p are the total number of configurations and the number of particles in each of them, respectively; $\Delta F_{jk}^{\mathrm{ref}}$ is the reference quantum deviation on particle j in configuration k that can be computed from a standard NMPIMD simulation; and $\Delta F_{jk}^{\mathrm{mod}}$ is the classical-like model of the quantum deviation which depends on M adjustable parameters, $a_1, \ldots, a_M$. Assuming that the forces between particles can be described as pairwise additive, $\Delta F_{jk}^{\mathrm{mod}}$ can be expressed as

$$\Delta F_{jk}^{\mathrm{mod}}(\{\mathbf{R}_j\}_k) = \sum_{i=1}^{N_p} f_{ij}^{\mathrm{mod}}(R_{ij})\mathbf{n}_{ij}, \tag{14}$$

where $\mathbf{n}_{ij} = \mathbf{R}_{ij}/|\mathbf{R}_{ij}|$, and $f_{ij}^{\mathrm{mod}}(R_{ij})$ are functions modeling the interparticle forces. Combining Eqs. (13) and (14), the least-squares minimization reduces in finding the solution of the following set of equations:

$$\Delta F_{jk}^{\mathrm{ref}}(\{\mathbf{R}_j\}_k) = \sum_{i=1}^{N_p} f_{ij}^{\mathrm{mod}}(R_{ij})\mathbf{n}_{ij}, \quad j = 1, \ldots, N_p; \quad k = 1, \ldots, N_c. \tag{15}$$

In the case that $N_p \times N_c$ is sufficiently large, these equations

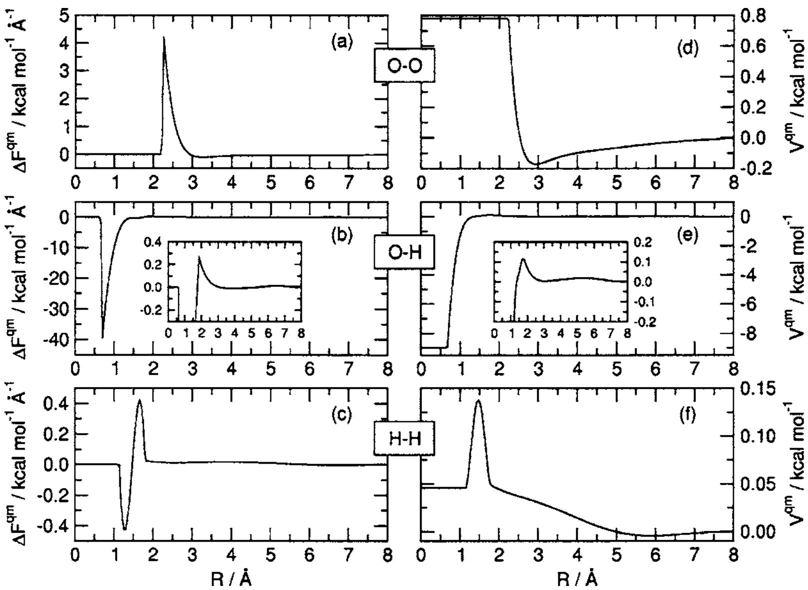

FIG. 6. Pair forces derived from the quantum deviation, Eq. (12), and corresponding potential curves for O–O [panels (a) and (d)], O–H [panels (b) and (e)], and H–H [panels (c) and (f)].

form an over determined set of linear equations that can be solved by using singular value decomposition.[44]

In order to generate the reference force F^c of Eq. (12), NMPIMD simulations with the q-SPC/Fw water model were performed in the *NVT* ensemble as illustrated in the previous section. The classical model for the quantum deviation, Eq. (14), was represented by cubic splines that describe the O–O, O–H, and H–H interactions. The use of cubic splines guarantees that the model force depends linearly on its parameters, and moreover it has smooth first derivative and continuous second derivative. In order to ensure that the set of linear equations in Eq. (15) was over determined, the number of molecular configurations was taken as $N_c=5$. The set of linear equations was solved for 2000 different configurations and the solutions were then averaged. The resulting effective forces $\Delta F^{\rm qm}$ derived from the quantum deviation, Eq. (12), and the corresponding potentials are shown in Fig. 6. For the O–O component, the potential assumes a simple form with both a (short-range) repulsive region and a (long-range) attractive region. It can be accurately fitted to a sum of an exponential function describing the short-range part, and a series of inverse power of the interatomic separation for the long-range part. It thus appears that for the O–O component the effects of the nuclear quantization can be described by an additional contribution to the Lennard-Jones potential from the classical force field. By contrast, the O–H and H–H potentials have more complicated forms since they include both intramolecular and intermolecular contributions. However, it is possible to note that the effects of the nuclear quantization are particularly important in the short-range (intramolecular) regions, while they are essentially negligible at a large distance. This is due to the inclusion of the zero-point energy that, as expected, is especially large for the intramolecular vibrations and is instead completely neglected in the classical force field. Tabulated values of $\Delta F^{\rm qm}$ are available from the authors upon request.

The q-SPC/Fw water model with the addition of $\Delta F^{\rm qm}$ was then employed in purely classical simulations, i.e., force-matched CMD simulations (FM-CMD) of liquid water at $T=298.15$ K and $P=1$ atm. The equilibrium properties were computed from a 1 ns *NVT* trajectory, while the dynamical properties were computed in the *NVE* ensemble averaging over 100 independent trajectories of 10 ps each. In all calculations a time step $\Delta t=0.5$ fs was employed.

B. Results

In Fig. 7 the radial distribution functions obtained from the FM-CMD simulations are compared to those obtained from full CMD simulations. It is important to note that the centroid radial distribution functions do not correspond to the true quantum RDF's shown in Fig. 4. As discussed in Ref. 45, a deconvolution procedure must be applied to any centroid RDF in order to be compared with both the usual quantum (obtained here from NMPIMD simulations) and experimental results. The FM-CMD radial distribution functions are found to be in essentially exact agreement with their full CMD analogs. Comparison between the velocity autocorrelation function computed with both the FM-CMD and the full CMD approaches is shown in Fig. 8. Also in this case a nearly perfect agreement is found. The values for several equilibrium and dynamical quantities computed with the FM-CMD method are listed in Table III. Excellent agreement with the corresponding results obtained from full CMD simulations (last column of Table II) is achieved for all the dynamical properties. Small differences are instead found for both ε and $C_V(T)$. This level of agreement was expected given the small differences between the CMD (and consequently FM-CMD) and NMPIMD radial distribution functions. However, it is important to note that all the quantum equilibrium properties can be computed during the NMPIMD simulation that is necessarily required by the FM-CMD method for the predetermination of the centroid force.

IV. CONCLUSIONS

In this study a detailed analysis of the effects of nuclear quantization on the equilibrium and dynamical properties of liquid water at ambient conditions has been carried out.

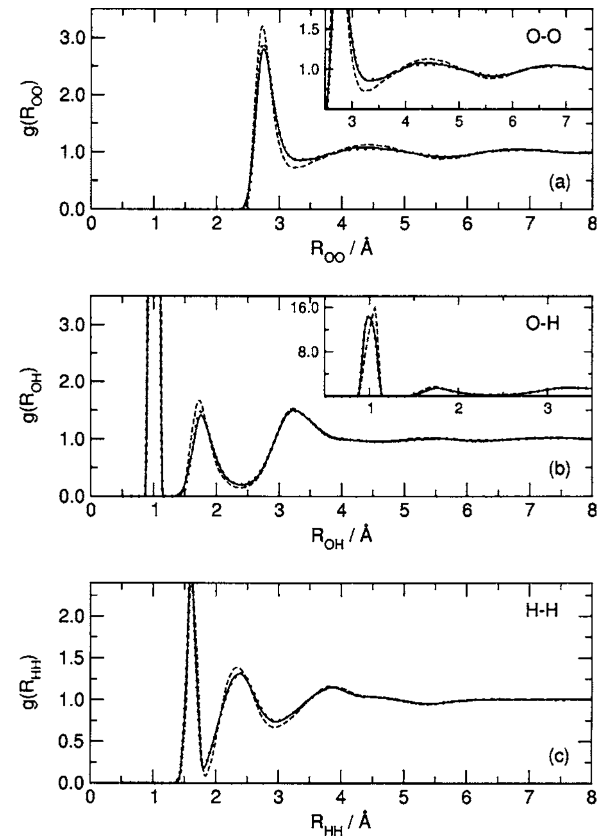

FIG. 7. Centroid radial distribution functions for oxygen-oxygen (a), oxygen-hydrogen (b), and hydrogen-hydrogen (c) atom pairs computed with the q-SPC/Fw water model. Solid line: centroid RDF's computed from force-matched centroid molecular dynamics (FM-CMD) simulations. Dotted line: centroid RDF's from full CMD simulations. Dashed line: results from classical molecular dynamics simulations.

Normal-mode path-integral molecular dynamics and centroid molecular dynamics simulations were performed with the recently developed SPC/Fw water model. Comparison between quantum and classical calculations indicates that inclusion of

TABLE III. Equilibrium and dynamical properties computed using force-matched centroid molecular dynamics (FM-CMD) simulations with the effective quantum model. The statistical uncertainty is given as standard deviation.

	FM-CMD q-SPC/Fw
ε	94 ± 2
C_V (cal mol^{-1} K^{-1})	25 ± 1
D (Å ps^{-1})	0.24 ± 0.01
τ_D (ps)	8.5 ± 0.5
τ_1^{HH} (ps)	3.9 ± 0.1
τ_2^{HH} (ps)	1.95 ± 0.05
τ_1^{OH} (ps)	4.2 ± 0.1
τ_2^{OH} (ps)	1.80 ± 0.05
τ_1^{μ} (ps)	4.8 ± 0.1
τ_2^{μ} (ps)	1.70 ± 0.05

the nuclear quantization results in a significantly less structured liquid, which shows faster translational and rotational dynamics.

Since a relatively poor agreement between quantum and experimental values was obtained when quantizing the classical SPC/Fw model, the original parameters were modified leading to a new and fully quantum water model, the q-SPC/Fw model. When utilized in quantum water simulations, the new model provides an accurate description of several properties of liquid water at ambient conditions. Given the accuracy of this simple, nonpolarizable, quantum model, the present results also raise a question on the nature of quantum effects in real water. In particular, it has been shown here that the effects of nuclear quantization are significant, but also that the underlying water model can be simply reparametrized to describe experimental results quite accurately. Therefore, simply quantizing a classical water model and then studying "quantum effects" appear to not be particularly fruitful. Indeed, it would seem (as always) that the accurate

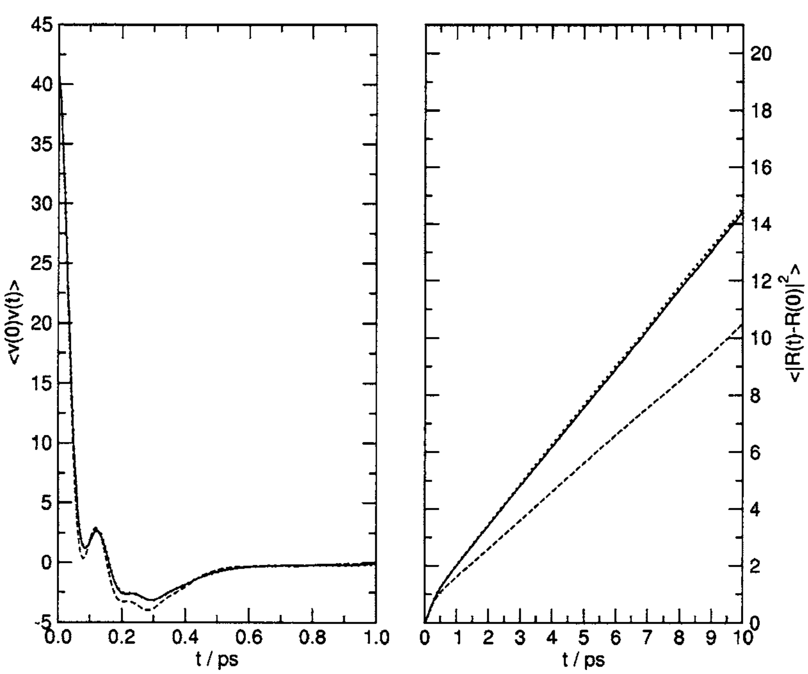

FIG. 8. Velocity autocorrelation function (left panel) and mean-square displacement (right panel) of liquid water computed with the q-SPC/Fw model. Solid line: results from force-matched centroid molecular dynamics (FM-CMD) simulations; dotted line: results from full centroid molecular dynamics simulations; dashed line: results from classical molecular dynamics simulations.

calculation and understanding of experimental observables such as isotope effects which may be related to underlying quantum effects is the best approach. It would also appear that the nuclear quantum effects may be as important as electronic polarization effects and should be concurrently treated when developing such polarizable models.

A force-matching approach within the CMD framework has also been successfully employed to derive an effective quantum model for liquid water in which the nuclear quantum effects are explicitly distinguished from those due to the underlying classical molecular interactions, and then independently modeled via classical-like forces. Excellent agreement is obtained between the results computed in standard classical simulations with this effective quantum model and those obtained from full CMD simulations with the pure q-SPC/Fw model. These results thus show that a force-matching approach within the CMD framework represents a computationally efficient and accurate method to perform quantum simulations of reasonably complex many-body systems such as liquid water.

The present q-SPC/Fw model is, to our knowledge, the first water model fully optimized using quantum simulations. Given its simplicity and accuracy, this new model can be combined with the common force fields available for proteins (possibly also reparametrized for quantum effects) to perform fully quantum biosimulations. Such simulations can help us investigate the role played by quantum mechanics in biomolecular systems as well as help reveal the properties and behavior of the water molecules that are located in the immediate vicinity of a biomolecule (“biological water”).

ACKNOWLEDGMENTS

This research was supported by the Office of Naval Research through Grant No. N00014-05-1-0457. One of the authors (F.P.) would like to thank Kim Wong, Takeshi Yamamoto, and Sergei Izvekov for many stimulating discussions.

[1] W. L. Jorgensen, J. Chandrasekhar, J. D. Madura, R. W. Impey, and M. L. Klein, J. Chem. Phys. **79**, 926 (1983); H. J. C. Berendsen, J. P. M. Postma, W. F. van Gunsteren, and J. Hermans, in *Intermolecular Forces*, edited by B. Pullman (Reidel, Dordrecht, 1981), p. 331; H. J. C. Berendsen, J. R. Grigera, and T. P. Straatsma, J. Phys. Chem. **91**, 6269 (1987); J. L. F. Abascal and C. Vega, J. Chem. Phys. **123**, 234505 (2005).
[2] H. W. Horn, W. C. Swope, J. W. Pitera, J. D. Madura, T. J. Dick, G. L. Hura, and T. Head-Gordon, J. Chem. Phys. **120**, 9665 (2004).
[3] L. X. Dang and B. M. Pettitt, J. Phys. Chem. **91**, 3349 (1987); K. Toukan and A. Rahman, Phys. Rev. B **31**, 2643 (1985).
[4] D. M. Ferguson, J. Comput. Chem. **16**, 501 (1995); O. Teleman, B. Jonsson, and S. Engstrom, Mol. Phys. **60**, 193 (1987); A. D. Trokhymchuk, M. F. Holovko, and K. Heinzinger, J. Chem. Phys. **99**, 2964 (1993).
[5] H. Saint-Martin, J. Hernandez-Cobos, M. I. Bernal-Uruchurtu, I. Ortega-Blake, and H. J. C. Berendsen, J. Chem. Phys. **113**, 10899 (2000).
[6] G. Lamoureux, A. D. J. MacKerell, and B. Roux, J. Chem. Phys. **119**, 5185 (2003); S. W. Rick, S. J. Stuart, and B. J. Berne, *ibid.* **101**, 6141 (1994); C. J. Burnham and S. S. Xantheas, *ibid.* **116**, 5115 (2002); J. Jeon, A. E. Lefohn, and G. A. Voth, *ibid.* **118**, 7504 (2003); S. Iuchi, A. Morita, and S. Kato, J. Phys. Chem. B **106**, 3466 (2002); P. Ren and J. W. Ponder, *ibid.* **107**, 5933 (2003).
[7] A. G. Donchev, V. D. Ozrin, M. V. Subbotin, O. V. Tarasov, and V. I. Tarasov, Proc. Natl. Acad. Sci. U.S.A. **102**, 7829 (2005).
[8] R. A. Kuharski and P. J. Rossky, J. Chem. Phys. **82**, 5164 (1985).
[9] J. Lobaugh and G. A. Voth, J. Chem. Phys. **106**, 2400 (1997).
[10] J. A. Poulsen, G. Nyman, and P. J. Rossky, Proc. Natl. Acad. Sci. U.S.A. **102**, 6709 (2005).
[11] A. Wallqvist and B. J. Berne, Chem. Phys. Lett. **117**, 214 (1985).
[12] B. Guillot and Y. Guissani, J. Chem. Phys. **108**, 10162 (1998).
[13] L. Hernández de la Peña and P. G. Kusalik, J. Chem. Phys. **121**, 5992 (2004); J. Am. Chem. Soc. **127**, 5246 (2005); J. Chem. Phys. **125**, 054512 (2006).
[14] T. F. Miller III and D. E. Manolopoulos, J. Chem. Phys. **123**, 154504 (2005).
[15] H. A. Stern and B. J. Berne, J. Chem. Phys. **115**, 7622 (2001).
[16] A. G. Donchev, N. G. Galkin, A. A. Illarionov, O. V. Khoruzhii, M. A. Olevanov, V. D. Ozrin, M. V. Subbotin, and V. I. Tarasov, Proc. Natl. Acad. Sci. U.S.A. **103**, 8613 (2006).
[17] D. A. Case, T. E. Cheatham III, T. A. Darden, H. Gohlke, R. Luo, K. M. Merz, Jr., A. Onufriev, C. Simmerling, B. Wang, and R. Woods, J. Comput. Chem. **26**, 1668 (2005); AMBER, version 9; http://amber.scripps.edu
[18] B. R. Brooks, R. E. Bruccoleri, B. D. Olafson, D. J. States, S. Swaminathan, and M. Karplus, J. Comput. Chem. **4**, 187 (1983); M. E. Tuckerman, D. A. Yarne, S. O. Samuelson, A. L. Hughes, and G. J. Martyna, Comput. Phys. Commun. **128**, 333 (2000).
[19] J. S. Cao and G. A. Voth, J. Chem. Phys. **100**, 5093 (1994); **100**, 5106 (1994); **101**, 6157 (1994); **101**, 6184 (1994).
[20] J. S. Cao and G. A. Voth, J. Chem. Phys. **101**, 6168 (1994).
[21] S. Jang and G. A. Voth, J. Chem. Phys. **111**, 2357 (1999); **111**, 2371 (1999).
[22] D. R. Reichman and E. Rabani, Phys. Rev. Lett. **87**, 265702 (2001); J. Chem. Phys. **116**, 6271 (2002).
[23] N. Makri and K. Thompson, Chem. Phys. Lett. **291**, 101 (1998); K. Thompson and N. Makri, J. Chem. Phys. **110**, 1343 (1999); X. Sun and W. H. Miller, *ibid.* **110**, 6635 (1999).
[24] J. A. Poulsen, G. Nyman, and P. J. Rossky, J. Chem. Phys. **119**, 12179 (2003).
[25] G. Krilov and B. J. Berne, J. Chem. Phys. **111**, 9140 (1999).
[26] T. D. Hone, S. Izvekov, and G. A. Voth, J. Chem. Phys. **122**, 054105 (2005).
[27] B. J. Berne and D. Thirumalai, Annu. Rev. Phys. Chem. **37**, 401 (1986); M. E. Tuckerman, in *Quantum Simulations of Complex Many-Body Systems: From Theory to Algorithms*, edited by J. Grotendorst, D. Marx, and A. Muramatsu (John von Neumann Institute for Computing, Jülich, 2002), Vol. 10, p. 269.
[28] Y. J. Wu, H. L. Tepper, and G. A. Voth, J. Chem. Phys. **124**, 024503 (2006).
[29] T. Darden, D. York, and L. Pedersen, J. Chem. Phys. **98**, 10089 (1993); U. Essmann, L. Perera, M. L. Berkowitz, T. Darden, H. Lee, and L. G. Pedersen, *ibid.* **103**, 8577 (1995); H. G. Petersen, *ibid.* **103**, 3668 (1995).
[30] H. J. C. Berendsen, J. P. M. Postma, W. F. van Gunsteren, A. Di Nola, and J. R. Haak, J. Chem. Phys. **81**, 3684 (1984).
[31] M. P. Allen and D. J. Tildesley, *Computer Simulations of Liquids* (Oxford University Press, Oxford, 1987).
[32] G. J. Martyna, M. L. Klein, and M. Tuckerman, J. Chem. Phys. **97**, 2635 (1992).
[33] G. J. Martyna, A. Hughes, and M. E. Tuckerman, J. Chem. Phys. **110**, 3275 (1999).
[34] D. R. Reichman, P.-N. Roy, S. Jang, and G. A. Voth, J. Chem. Phys. **113**, 919 (2000).
[35] A. K. Soper, Chem. Phys. **258**, 121 (2000).
[36] J. M. Sorenson, G. Hura, R. M. Glaeser, and T. Head-Gordon, J. Chem. Phys. **113**, 9149 (2000).
[37] M. Shiga and W. Shinoda, J. Chem. Phys. **123**, 134502 (2005); T. M. Yamamoto, *ibid.* **123**, 104101 (2005).
[38] M. F. Herman, E. J. Bruskin, and B. J. Berne, J. Chem. Phys. **76**, 5150 (1982).
[39] M. Neumann, J. Chem. Phys. **82**, 5663 (1985).
[40] R. Zwanzig, *Nonequilibrium Statistical Mechanics* (Oxford University Press, Oxford, 2001).
[41] D. Eisenberg and W. Kauzmann, *The Structure and Properties of Water* (Clarendon, Oxford, 1969).
[42] J. E. Bertie and Z. Lan, Appl. Spectrosc. **50**, 1047 (1996).
[43] B. Guillot, J. Chem. Phys. **95**, 1543 (1991).
[44] S. Izvekov, M. Parrinello, C. J. Burnham, and G. A. Voth, J. Chem. Phys. **120**, 10896 (2004).
[45] N. Blinov and P.-N. Roy, J. Chem. Phys. **120**, 3759 (2004).